

This is a copy of a conversation between ChatGPT & Anonymous.

Report conversation

synovial cell SubClassOf Nothing
synovial cell SubClassOf part of some synovial joint
synovial joint SubClassOf surrounded by some articular capsule
articular capsule SubClassOf has part some layer of synovial tissue
layer of synovial tissue EquivalentTo serous membrane and (produces some synovial fluid)
synovial fluid EquivalentTo transudate and (produced by some synovial cell)
transudate EquivalentTo organism substance and (has quality some quality of a liquid) and (transformation of some blood plasma) and (filtered_through some capillary)
transformation of SubPropertyOf: develops from
develops from SubPropertyOf: has developmental contribution from
has developmental contribution from Domain anatomical entity
anatomical entity SubClassOf material entity
capillary SubClassOf connects some arteriole
arteriole SubClassOf connects some artery
artery SubClassOf arterial blood vessel
arterial blood vessel EquivalentTo blood vessel and (part of some arterial system)
arterial system SubClassOf vascular system
vascular system SubClassOf part of some cardiovascular system
cardiovascular system SubClassOf has part some heart
heart SubClassOf part of some heart plus pericardium
heart plus pericardium SubClassOf thoracic cavity element
thoracic cavity element EquivalentTo organ and (located in some thoracic cavity)
thoracic cavity EquivalentTo anatomical space and (part of some coeleomic cavity lumen) and (luminal space of some

thoracic segment of trunk)
luminal space of Domain immaterial entity
coelemic cavity lumen SubClassOf transformation of
some future coelemic cavity lumen
coelemic cavity lumen SubClassOf luminal space of some
coelom
material entity DisjointWith immaterial entity
epithelial cell of lung SubClassOf Nothing
epithelial cell of lung SubClassOf part of some lung
lung SubClassOf thoracic cavity element
thoracic cavity element SubClassOf located in some
thoracic cavity
thoracic cavity EquivalentTo anatomical space and (part of
some coelemic cavity lumen) and (luminal space of some
thoracic segment of trunk)
coelemic cavity lumen SubClassOf transformation of
some future coelemic cavity lumen
transformation of SubPropertyOf: develops from
develops from SubPropertyOf: has developmental
contribution from
has developmental contribution from Domain anatomical
entity
anatomical entity SubClassOf material entity
luminal space of Domain immaterial entity
coelemic cavity lumen SubClassOf luminal space of some
coelom
material entity DisjointWith immaterial entity
club cell SubClassOf Nothing
club cell SubClassOf epithelial cell of tracheobronchial
tree
epithelial cell of tracheobronchial tree SubClassOf
epithelial cell of lower respiratory tract
epithelial cell of lower respiratory tract SubClassOf part
of some lower respiratory tract
lower respiratory tract SubClassOf has part some pair of
lungs
pair of lungs SubClassOf located in some thoracic cavity
thoracic cavity EquivalentTo anatomical space and (part of
some coelemic cavity lumen) and (luminal space of some
thoracic segment of trunk)

coelemic cavity lumen SubClassOf transformation of
some future coelemic cavity lumen
transformation of SubPropertyOf: develops from
develops from SubPropertyOf: has developmental
contribution from
has developmental contribution from Domain anatomical
entity
anatomical entity SubClassOf material entity
luminal space of Domain immaterial entity
coelemic cavity lumen SubClassOf luminal space of some
coelom
material entity DisjointWith immaterial entity
luteal cell SubClassOf Nothing
luteal cell SubClassOf part of some corpus luteum
corpus luteum SubClassOf develops from some ovarian
follicle
develops from SubPropertyOf: has developmental
contribution from
has developmental contribution from Domain anatomical
entity
anatomical entity SubClassOf material entity
ovarian follicle SubClassOf develops from some ovary sex
cord
ovary sex cord SubClassOf develops from some primitive
sex cord of indifferent gonad
primitive sex cord of indifferent gonad SubClassOf
develops from some coelomic epithelium
coelomic epithelium SubClassOf located in some coelemic
cavity lumen
coelemic cavity lumen SubClassOf transformation of
some future coelemic cavity lumen
transformation of SubPropertyOf: develops from
coelemic cavity lumen SubClassOf luminal space of some
coelom
luminal space of Domain immaterial entity
material entity DisjointWith immaterial entity
epithelial cell of pancreas SubClassOf Nothing
epithelial cell of pancreas SubClassOf part of some
pancreas
pancreas SubClassOf viscus

viscus EquivalentTo organ and (located in some coelemic cavity lumen)

coelemic cavity lumen SubClassOf transformation of some future coelemic cavity lumen

transformation of SubPropertyOf: develops from

develops from SubPropertyOf: has developmental contribution from

has developmental contribution from Domain anatomical entity

anatomical entity SubClassOf material entity

coelemic cavity lumen SubClassOf luminal space of some coelom

luminal space of Domain immaterial entity

material entity DisjointWith immaterial entity

type B pancreatic cell SubClassOf Nothing

type B pancreatic cell EquivalentTo enteroendocrine cell and (part of some islet of Langerhans) and (capable of some insulin secretion)

islet of Langerhans SubClassOf contributes to morphology of some endocrine pancreas

endocrine pancreas SubClassOf contributes to morphology of some pancreas

pancreas SubClassOf has developmental contribution from some ventral pancreatic bud

has developmental contribution from Domain anatomical entity

anatomical entity SubClassOf material entity

ventral pancreatic bud SubClassOf develops from some hepatic diverticulum

hepatic diverticulum SubClassOf part of some septum transversum

septum transversum SubClassOf located in some coelemic cavity lumen

coelemic cavity lumen SubClassOf transformation of some future coelemic cavity lumen

transformation of SubPropertyOf: develops from

coelemic cavity lumen SubClassOf luminal space of some coelom

luminal space of Domain immaterial entity

develops from SubPropertyOf: has developmental

contribution from
material entity DisjointWith immaterial entity
pancreatic A cell SubClassOf Nothing
pancreatic A cell EquivalentTo type A enteroendocrine
cell and (part of some pancreas)
pancreas SubClassOf viscus
viscus EquivalentTo organ and (located in some coelemic
cavity lumen)
coelemic cavity lumen SubClassOf transformation of
some future coelemic cavity lumen
transformation of SubPropertyOf: develops from
develops from SubPropertyOf: has developmental
contribution from
has developmental contribution from Domain anatomical
entity
anatomical entity SubClassOf material entity
coelemic cavity lumen SubClassOf luminal space of some
coelom
luminal space of Domain immaterial entity
material entity DisjointWith immaterial entity
hepatocyte SubClassOf Nothing
hepatocyte SubClassOf part of some liver
liver SubClassOf develops from some septum
transversum
develops from SubPropertyOf: has developmental
contribution from
has developmental contribution from Domain anatomical
entity
anatomical entity SubClassOf material entity
septum transversum SubClassOf located in some
coelemic cavity lumen
coelemic cavity lumen SubClassOf transformation of
some future coelemic cavity lumen
transformation of SubPropertyOf: develops from
coelemic cavity lumen SubClassOf luminal space of some
coelom
luminal space of Domain immaterial entity
material entity DisjointWith immaterial entity
blood vessel endothelial cell SubClassOf Nothing
blood vessel endothelial cell SubClassOf part of some

blood vessel endothelium
blood vessel endothelium EquivalentTo endothelium and
(part of some blood vessel)
blood vessel SubClassOf channel_for some blood
blood SubClassOf located in some vasculature
vasculature SubClassOf part of some vascular system
vascular system SubClassOf part of some cardiovascular
system
cardiovascular system SubClassOf has part some heart
heart SubClassOf part of some heart plus pericardium
heart plus pericardium SubClassOf thoracic cavity
element
thoracic cavity element EquivalentTo organ and (located
in some thoracic cavity)
thoracic cavity SubClassOf part of some coelemic cavity
lumen
coelemic cavity lumen SubClassOf transformation of
some future coelemic cavity lumen
transformation of SubPropertyOf: develops from
develops from SubPropertyOf: has developmental
contribution from
has developmental contribution from Domain anatomical
entity
anatomical entity SubClassOf material entity
coelemic cavity lumen SubClassOf luminal space of some
coelom
luminal space of Domain immaterial entity
Reflexive: has part
immaterial entity DisjointWith has part some material
entity
pancreatic D cell SubClassOf Nothing
pancreatic D cell SubClassOf pancreatic endocrine cell
pancreatic endocrine cell EquivalentTo endocrine cell and
(part of some pancreas)
pancreas SubClassOf viscus
viscus EquivalentTo organ and (located in some coelemic
cavity lumen)
coelemic cavity lumen SubClassOf transformation of
some future coelemic cavity lumen
transformation of SubPropertyOf: develops from

develops from SubPropertyOf: has developmental contribution from
has developmental contribution from Domain anatomical entity
anatomical entity SubClassOf material entity
coelemic cavity lumen SubClassOf luminal space of some coelom
luminal space of Domain immaterial entity
material entity DisjointWith immaterial entity
Axiom Impact
Axioms used 10 times
anatomical entity SubClassOf material entity
[foodon_import.owl]
coelemic cavity lumen SubClassOf transformation of some future coelemic cavity lumen [uberon_import.owl]
coelemic cavity lumen SubClassOf luminal space of some coelom [uberon_import.owl]
develops from SubPropertyOf: has developmental contribution from [maxo_import.owl]
transformation of SubPropertyOf: develops from [ro_import.owl,envo_import.owl]
has developmental contribution from Domain anatomical entity [ecto_import.owl,envo_import.owl]
luminal space of Domain immaterial entity [ro_import.owl]
Axioms used 9 times
material entity DisjointWith immaterial entity [ro_import.owl,envo_import.owl]
Axioms used 3 times
viscus EquivalentTo organ and (located in some coelemic cavity lumen) [uberon_import.owl]
thoracic cavity EquivalentTo anatomical space and (part of some coelemic cavity lumen) and (luminal space of some thoracic segment of trunk) [uberon_import.owl]
pancreas SubClassOf viscus [uberon_import.owl]
Axioms used 2 times
thoracic cavity element EquivalentTo organ and (located in some thoracic cavity) [uberon_import.owl]
heart SubClassOf part of some heart plus pericardium [uberon_import.owl]

septum transversum SubClassOf located in some coeleomic cavity lumen [uberon_import.owl]
 cardiovascular system SubClassOf has part some heart [uberon_import.owl]
 vascular system SubClassOf part of some cardiovascular system [uberon_import.owl]
 heart plus pericardium SubClassOf thoracic cavity element [uberon_import.owl]
 Axioms used 1 times
 type B pancreatic cell EquivalentTo enteroendocrine cell and (part of some islet of Langerhans) and (capable of some insulin secretion) [cl_import.owl]
 pancreatic A cell EquivalentTo type A enteroendocrine cell and (part of some pancreas) [cl_import.owl]
 pancreatic endocrine cell EquivalentTo endocrine cell and (part of some pancreas) [cl_import.owl]
 synovial fluid EquivalentTo transudate and (produced by some synovial cell) [uberon_import.owl]
 arterial blood vessel EquivalentTo blood vessel and (part of some arterial system) [uberon_import.owl]
 blood vessel endothelium EquivalentTo endothelium and (part of some blood vessel) [uberon_import.owl]
 layer of synovial tissue EquivalentTo serous membrane and (produces some synovial fluid) [uberon_import.owl]
 transudate EquivalentTo organism substance and (has quality some quality of a liquid) and (transformation of some blood plasma) and (filtered_through some capillary) [uberon_import.owl]
 blood vessel endothelial cell SubClassOf part of some blood vessel endothelium [cl_import.owl]
 epithelial cell of lung SubClassOf part of some lung [cl_import.owl]
 epithelial cell of pancreas SubClassOf part of some pancreas [cl_import.owl]
 club cell SubClassOf epithelial cell of tracheobronchial tree [cl_import.owl]
 pancreatic D cell SubClassOf pancreatic endocrine cell

≡ ChatGPT 5.2 ✓

[cl_import.owl]

hepatocyte SubClassOf part of some liver [cl_import.owl]

synovial cell SubClassOf part of some synovial joint
[cl_import.owl]

epithelial cell of tracheobronchial tree SubClassOf
epithelial cell of lower respiratory tract [cl_import.owl]
epithelial cell of lower respiratory tract SubClassOf part
of some lower respiratory tract [cl_import.owl]

islet of Langerhans SubClassOf contributes to
morphology of some endocrine pancreas
[uberon_import.owl]

endocrine pancreas SubClassOf contributes to
morphology of some pancreas [uberon_import.owl]
pair of lungs SubClassOf located in some thoracic cavity
[uberon_import.owl]

blood SubClassOf located in some vasculature
[uberon_import.owl]

pancreas SubClassOf has developmental contribution
from some ventral pancreatic bud [uberon_import.owl]

ovarian follicle SubClassOf develops from some ovary sex
cord [uberon_import.owl]

articular capsule SubClassOf has part some layer of
synovial tissue [uberon_import.owl]

lower respiratory tract SubClassOf has part some pair of
lungs [uberon_import.owl]

artery SubClassOf arterial blood vessel
[uberon_import.owl]

arteriole SubClassOf connects some artery
[uberon_import.owl]

blood vessel SubClassOf channel_for some blood
[uberon_import.owl]

capillary SubClassOf connects some arteriole
[uberon_import.owl]

lung SubClassOf thoracic cavity element
[uberon_import.owl]

vasculature SubClassOf part of some vascular system
[uberon_import.owl]

liver SubClassOf develops from some septum
transversum [uberon_import.owl]

synovial joint SubClassOf surrounded by some articular
capsule [uberon_import.owl]

thoracic cavity SubClassOf part of some coeleomic cavity

lumen [uberon_import.owl]

corpus luteum SubClassOf develops from some ovarian

follicle [uberon_import.owl]

ventral pancreatic bud SubClassOf develops from some

hepatic diverticulum [uberon_import.owl]

arterial system SubClassOf vascular system

[uberon_import.owl]

thoracic cavity element SubClassOf located in some

thoracic cavity [uberon_import.owl]

ovary sex cord SubClassOf develops from some primitive

sex cord of indifferent gonad [uberon_import.owl]

coelomic epithelium SubClassOf located in some coeleomic

cavity lumen [uberon_import.owl]

hepatic diverticulum SubClassOf part of some septum

transversum [uberon_import.owl]

primitive sex cord of indifferent gonad SubClassOf

develops from some coelomic epithelium

[uberon_import.owl]

immaterial entity DisjointWith has part some material

entity [ro_import.owl,envo_import.owl]

Reflexive: has part [ecto_import.owl,foodon_import.owl]

Ontologies used:

foodon_import.owl

(http://purl.obolibrary.org/obo/mondo/imports/foodon_import.owl)

ecto_import.owl

(http://purl.obolibrary.org/obo/mondo/imports/ecto_import.owl)

cl_import.owl

(http://purl.obolibrary.org/obo/mondo/imports/cl_import.owl)

envo_import.owl

(http://purl.obolibrary.org/obo/mondo/imports/envo_import.owl)

maxo_import.owl

(http://purl.obolibrary.org/obo/mondo/imports/maxo_import.owl)

ro_import.owl

(http://purl.obolibrary.org/obo/mondo/imports/ro_import.owl)

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t.owl)  
uberon_import.owl  
(http://purl.obolibrary.org/obo/mondo/imports/uberon_i  
mport.owl)  
@sabrinatoro
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The information you've provided is a detailed ontology or logical framework that describes anatomical entities and their relationships, largely in terms of development, composition, and spatial arrangement. The ontology uses formal relationships like "

It appears that you've provided a set of axioms and relationships pertaining to biological entities, especially anatomical structures and cells. This looks like a part of an ontology, which is a formal way to describe types, properties, and interrelationships of entities in a specific domain—in this case, likely biology or anatomy.

However, there seems to be a recurring logical issue in your ontology, particularly related to the term "coelemic cavity lumen". You have "coelemic cavity lumen" as a subclass of "luminal space of some coelom," and "luminal space of" is in the domain of "immaterial entity." However, "anatomical entity" is a subclass of "material entity," and "material entity" is disjoint with "immaterial entity."

This logical conflict might cause inconsistency issues in your ontology. You'd likely need to resolve this to ensure that all your terms are logically coherent. The problem seems to arise multiple times, so addressing it could be crucial for the reliability and applicability of the ontology.

Would you like to discuss ways to resolve this issue or any other specific aspects of the ontology you're working on?